

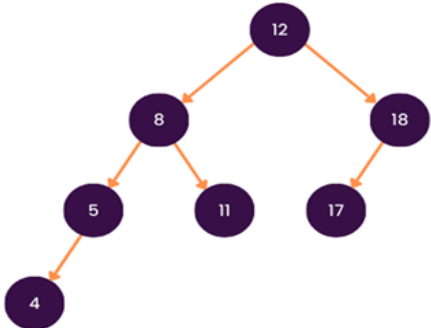
Assignment Title: (CO4-PO1)
Understanding the function and use of Tree data structure.

Understanding: Design, Application & Reflection

The Understanding with **Application & Reflection** tailored to the **Data Structure (CO4–PO1)** assignment — Understanding with applications and designed to connect theoretical understanding with practical, real-world use:

Objective:
To use the concepts of data structures with real-life applications and design the particular data structures for understanding of how appropriate data structure selection improves system efficiency.

Task: Understand the Tree Concept:



- For the above Tree data structure, answer the following questions:
- (i) Write the **all possible names** of the tree and hence explain (with proof) why they are called so.
 - (ii) Draw **max heap and min heap**.
 - (iii) Write **one real world application** for each of them.
 - (iv) Analysis the **time complexity** for each of them.
 - (v) Which one is the **best** among them and why?
 - (vi) Insert a node 10 using **Hand and Code** for each of them as answered the name of trees in (i).
 - (vii) Delete a node 8 using **Hand and Code** for each of them as answered the name of trees in (i).
 - (viii) Draw the memory representation and Linked List representation.
 - (ix) Traverse the tree in in order, preorder and post order using **Stack**.
 - (x) Draw the m-array trees as you wish.

Evaluation (5 Marks — Application & Reflection Component):

Criteria	Description	Marks
Understanding and Application of Tree	Correct understand, application and justification	2
Complexity Analysis to understand application accuracy	Understanding correctly for applications	1
Reflection Quality	Depth of Design understanding and clarity of expression	3
	Total	5

Guidelines

- 1. Complete all questions individually and neatly.
- 2. Include sections: Cover Page, Objectives, Background, Algorithm/Flowchart (where applicable), Application & Reflection, and Conclusion.
- 3. Submit clearly handwritten Report.
- 4. Plagiarism or copied content will not be accepted.
- 5. Follow professional formatting and label all diagrams clearly.
- 6. Submission Deadline: within 25 April, 2026, Time: upto 1:20 PM